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Short Report

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A Robust Numerical Framework for Ultrasonic Wave Characterization in Complex Petrophysical Environments

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Abstract

Characterizing rock heterogeneities remains a fundamental challenge in petrophysics. This study presents a k-space pseudospectral framework (k-Wave) to model 1 MHz ultrasonic wave propagation in a heterogeneous matrix ($v_f = 2500$ m/s) with circular inclusions ($v_s = 350$ m/s). Steady-state analysis using a 16384-point FFT revealed significant scattering, wavefront distortions, and acoustic shadow zones. Quantitative assessment using Contrast Ratio (CR), SNR, and effective attenuation (α_{eff}) provided precise structural characterization. The integration of the Convergent Born Series (CBS) and fractional viscoacoustic modeling ensured numerical stability and accounted for power-law absorption. Crucially, a comparative analysis showed only a 1% numerical discrepancy between the k-Wave and Modified Born Approach (MBA) results, validating the framework's accuracy for advanced petrophysical applications.

Keywords: Petrophysics, Ultrasonic Testing, k-Wave Simulation, Scattering Potential, Rock Heterogeneity, k-space Pseudospectral Method.

1. Introduction

Ultrasonic wave propagation serves as a critical diagnostic tool in petrophysics for characterizing the internal micro-geometry and fluid saturation states of reservoir rocks [4,6,7]. Recent advancements in numerical modeling have shifted towards k-space pseudospectral methods, such as the k-Wave framework, due to their superior ability to handle high-frequency wave propagation in complex, heterogeneous media with minimal numerical dispersion and computational cost [1].

In a petrophysical context, the interaction between ultrasonic waves and rock heterogeneity—primarily pores, micro-fractures, and mineral inclusions—triggers significant scattering and attenuation [3,5]. Research indicates that attenuation coefficients and velocity dispersion are strongly frequency-dependent, governed by mechanisms such as wave-induced fluid flow (squirt-flow) and matrix anelasticity at the core scale [2,4]. Furthermore, the relationship between ultrasonic velocity and petrophysical parameters like porosity and inclusion density remains a cornerstone of rock physics. Studies on shale and carbonate rocks demonstrate that velocity is inversely proportional to porosity and highly sensitive to the aspect ratio of micro-cracks, necessitating the use of sophisticated simulation models to invert these properties accurately [3].

However, a significant research gap exists in the seamless integration of high-resolution k-Wave simulations with multi-scale experimental laboratory data. Current literature emphasizes that numerical results, such as contrast ratios and effective attenuation coefficients, must be rigorously validated against semi-analytical methods like the modified Born

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approach or the Convergent Born Series (CBS) to ensure the reliability of petrophysical interpretations.

2. Materials and Methods

2.1. Generation of Thermoacoustic wave equation

Under the assumptions of lossless, homogeneous, and isotropic medium, the propagation of pressure field is governed by the classical acoustic wave equation. For a compressible fluid with density ρ, and the particle velocity $\mathbf{v}$, the conservation of mass (continuity equation) can be written as,

$$\partial_t \rho + \rho_0 \nabla . \mathbf{v} = 0, \tag{1}$$

To analyze small acoustic perturbations $\rho = \rho_0 + \rho_1$ and substituting in Eq. 1 and neglecting higher order nonlinear terms,

$$\partial_t \rho_1 + \rho_0 \nabla . \mathbf{v_1} = 0, \tag{2}$$

Using the linear acoustic relation between pressure (p_1) and density (ρ_1) $p = c^2 \rho$ and can rewrite the equation as,

$$\partial_t \rho_1 + \rho_0 c^2 \nabla . \mathbf{v_1} = 0, \tag{3}$$

From conservation of momentum and taking divergence on both sides,

$$\rho_0 \partial_t (\nabla . \mathbf{v_1}) = -\nabla^2 p, \tag{4}$$

Substituting all terms and from Eq. (1) the classical wave equation becomes,

$$\{\nabla^2 - (1/c^2)\partial_t^2\} p(\mathbf{r}, t) = 0, \tag{5}$$

If ultrasonic source term contribute,

$$\{\nabla^2 - (1/c^2)\partial_t^2\} p(\mathbf{r}, t) = -S(\mathbf{r}, t), \tag{6}$$

where C_p is the specific heat capacity at constant pressure, c is the speed of sound of the ambient medium.

2.2. Time domain pseudospectral approach

The second-order formulations, k-Wave utilizes a system of coupled first-order partial differential equations to model the physical dynamics of the medium. These governing equations, which account for mass and momentum conservation along with the equation of state, are expressed as follows [8]:

$$\frac{\partial \mathbf{u}}{\partial t} = -\frac{1}{\rho_f} \nabla \tilde{p} + S_F, \tag{7a}$$

$$\frac{\partial \rho}{\partial t} = -(2\rho + \rho_f) \nabla \cdot \mathbf{u} - \mathbf{u} \cdot \nabla \rho_f + S_M, \tag{7b}$$

$$\tilde{p} = c_f^2 \left(\rho + \mathbf{d} \cdot \nabla + \frac{B}{2A} \frac{\rho^2}{\rho_f} \right) - L_{\text{frac}} \tag{7c}$$

In Eq. (7c), the fractional Laplacian operator, L_{frac}, accounts for power-law absorption and is defined as $L_{\text{frac}} = \tau \frac{\partial}{\partial t} (-\nabla^2)^{\frac{\gamma}{2}-1} + \eta (-\nabla^2)^{\frac{\gamma+1}{2}-1}$. Here, $\mathbf{u}$ represents the particle velocity vector, and ρ denotes the acoustic density. The terms S_F and S_M correspond to body force and mass source inputs, respectively. In the present study, the mass source term was prioritized while the force source term was neglected. This pseudospectral approach computes spatial gradients via Fourier collocation, while the temporal evolution

is handled through a dispersion-corrected finite-difference scheme. For enhanced stability and accuracy in heterogeneous simulations, a staggered grid in both space and time is utilized.

2.3. Modified Born approach

To account for the heterogeneities within the rock matrix, we represent the discrete inclusions (e.g., pores or vugs) as a scattering potential, $V(\mathbf{r})$, which characterizes the contrast in acoustic properties between the host medium and the inclusions. The scattering potential is defined as $V(\mathbf{r}) = k_s^2 - k_f^2 - i\epsilon$, where k_s and k_f denote the wavenumbers associated with the inclusion and the surrounding fluid medium, respectively. The term ϵ is a real positive constant representing the intrinsic attenuation or energy dissipation within the medium. Under these considerations, the governing inhomogeneous Helmholtz equation for the acoustic pressure field $p(\mathbf{r})$ is formulated as:

$$\nabla^2 p(\mathbf{r}) + (k_f^2 + i\epsilon)p(\mathbf{r}) = -S(\mathbf{r}) - V(\mathbf{r})p(\mathbf{r}) \tag{8}$$

where $S(\mathbf{r})$ represents the source distribution. Equation (8) effectively describes the total pressure field resulting from the interaction between the incident wave and the scattering potential of the rock's internal structures. The choice of the artificial attenuation parameter ϵ is critical and depends on the magnitude of the medium's inhomogeneity. The solution to the governing wave equation can be expressed as a volume integral [9]:

$$p(\mathbf{r}) = \int g(\mathbf{r}|\mathbf{r}_0)[V(\mathbf{r}_0)p(\mathbf{r}_0) + S(\mathbf{r}_0)]d^3\mathbf{r}_0 \tag{9}$$

where the integration spans the entire spatial domain $\mathbb{R}^2$ or $\mathbb{R}^3$. The term $g(\mathbf{r}|\mathbf{r}_0)$ represents the Green's function, satisfying the following relationship:

$$\nabla^2 g(\mathbf{r}|\mathbf{r}_0) + (k_f^2 + i\epsilon)g(\mathbf{r}|\mathbf{r}_0) = -\delta(\mathbf{r} - \mathbf{r}_0) \tag{10}$$

Here, $\delta(\mathbf{r} - \mathbf{r}_0)$ is the Dirac delta function. where $M = \Gamma G V - \Gamma + I$. The solution can then be expanded into an infinite series:

$$p = \sum_{n=0}^{\infty} M^n \Gamma G S \tag{11}$$

This series converges provided that the spectral radius of M is less than unity ($|M| < 1$). Previous studies [9] have demonstrated that convergence is guaranteed for all structures if we choose $\Gamma = \frac{i}{\epsilon} V(\mathbf{r})$ and $\epsilon \geq \max(|k_s^2 - k_f^2|)$.

In realistic petrophysical scenarios, ultrasonic energy absorption often follows a power-law frequency dependence. To incorporate these effects, we utilize the fractional viscoacoustic wave equation proposed by Chen and Holm [8,10]:

$$\nabla^2 - v_f^{-2}\partial_t^2 + \tau\partial_t (-\nabla^2)^{\frac{\nu}{2}} + \eta(-\nabla^2)^{\frac{\nu+1}{2}}, p(\mathbf{r}, t) = 0 \tag{12}$$

where $\tau = -2\alpha_0 v_f^{\nu-1}$ and $\eta = 2\alpha_0 v_f^{\nu} \tan(\pi\nu/2)$ are constants related to power-law absorption and dispersion, respectively. In the frequency domain, this can be integrated into the CBS framework as:

$$[\nabla^2 + (k_f^2 + i\epsilon)]p(\mathbf{r}) = -S(\mathbf{r}) - -i\tau\omega (-\nabla^2)^{\frac{\nu}{2}} + \eta(-\nabla^2)^{\frac{\nu+1}{2}} + k_s^2 - k_f^2 - i\epsilon, p(\mathbf{r}) \tag{13}$$

3. Numerical Experiments and Performance Metrics

To demonstrate the efficacy of the proposed k-space pseudospectral framework in characterizing rock heterogeneities, a series of 2D numerical experiments were conducted. The simulation setup was specifically designed to replicate an ultrasonic testing environment where wave-matter interactions occur within a fluid-saturated rock matrix containing multiple inclusions.

3.1. Computational Grid and Medium Configuration

The simulation was implemented on a discrete computational domain of 128×128 grid points. The spatial resolution was set to $dx = dy = 0.16$ mm ($8 \times 0.02 \times 10^{-3}$ m), ensuring sufficient sampling of the ultrasonic wavelength. To eliminate unwanted boundary reflections, a Perfectly Matched Layer (PML) with a thickness of 10 grid points was integrated into the domain.

The ambient host medium (representing the fluid-filled matrix) was characterized by a longitudinal velocity of $v_f = 2500$ m/s and a mass density of $\rho_f = 1000$ kg/m^3. The rock's internal heterogeneities were modeled using five discrete circular inclusions (scatterers) with radii varying between 8 and 15 grid points. These inclusions were assigned a significantly lower propagation velocity of $v_s = 350$ m/s to simulate highly compliant features or vugs. To incorporate energy dissipation effects, a frequency-dependent absorption model based on Stokes' mode was applied, utilizing an absorption coefficient $\alpha_{\text{coeff}} = 0.75$ dB/(MHz$^y \cdot$ cm) and a power-law exponent $\alpha_{\text{power}} = 1.5$.

3.2. Source Excitation and Signal Acquisition

A linear ultrasonic source array consisting of 51 elements was positioned at the 12th grid location along the X-axis. The source emitted a continuous sinusoidal signal with a fundamental frequency $f = 1.007$ MHz and a peak pressure magnitude of 1 MPa (10^6 Pa). The temporal resolution was automatically adjusted to satisfy the Courant–Friedrichs–Lewy (CFL) stability criterion, fixed at 0.3 for all simulations.

The pressure field data were recorded over a duration corresponding to three full wave periods. To extract steady-state characteristics with high spectral fidelity, the recorded signals underwent a 16384-point Fast Fourier Transform (FFT). This allowed for the generation of high-resolution spatial maps of the steady-state amplitude (A_{ss}) and phase (Φ_{ss}) at the fundamental frequency.

3.3. Quantitative Metrics for Petrophysical Analysis

To objectively quantify the interaction between the ultrasonic waves and the simulated rock structures, several performance metrics were computed:

1. **Contrast Ratio (CR):** The acoustic contrast between the background matrix and the inclusions was evaluated in decibels (dB) as:

$$\text{CR} = 20 \log_{10} \left(\frac{\bar{A}_{\text{background}}}{\bar{A}_{\text{scatterer}}} \right) \tag{14}$$

   where $\bar{A}_{\text{background}}$ and $\bar{A}_{\text{scatterer}}$ represent the mean steady-state amplitudes in the fluid region and inclusion regions, respectively.

2. **Effective Attenuation Coefficient (α_{eff}):** Assuming an exponential decay model, the attenuation induced by the inclusions was estimated as:

$$\alpha_{\text{eff}} = -\frac{1}{d} \ln \left(\frac{\bar{A}_{\text{scatterer}}}{\bar{A}_{\text{background}}} \right) \tag{15}$$

where d is the Euclidean distance from the source center to the inclusion's centroid ($d \approx 0.0051$ m in the current setup).

3. **Signal-to-Noise Ratio (SNR):** The spatial stability and coherence of the background pressure field were quantified using the SNR:

$$\mathrm{SNR} = 20 \log_{10} \frac{\bar{A}_{\mathrm{background}}}{\sigma_{\mathrm{background}}} \quad (16)$$

where $\sigma_{\mathrm{background}}$ is the standard deviation of the amplitude within the background region.

4. **Transmission Coefficient (T):** The energy efficiency of the wave propagating through the scatterer cluster into the shadow zone was calculated as:

$$T = \frac{\bar{A}_{\mathrm{shadow}}}{\bar{A}_{\mathrm{background}}} \quad (17)$$

where $\bar{A}_{\mathrm{shadow}}$ represents the mean amplitude in the region distal to the inclusions.

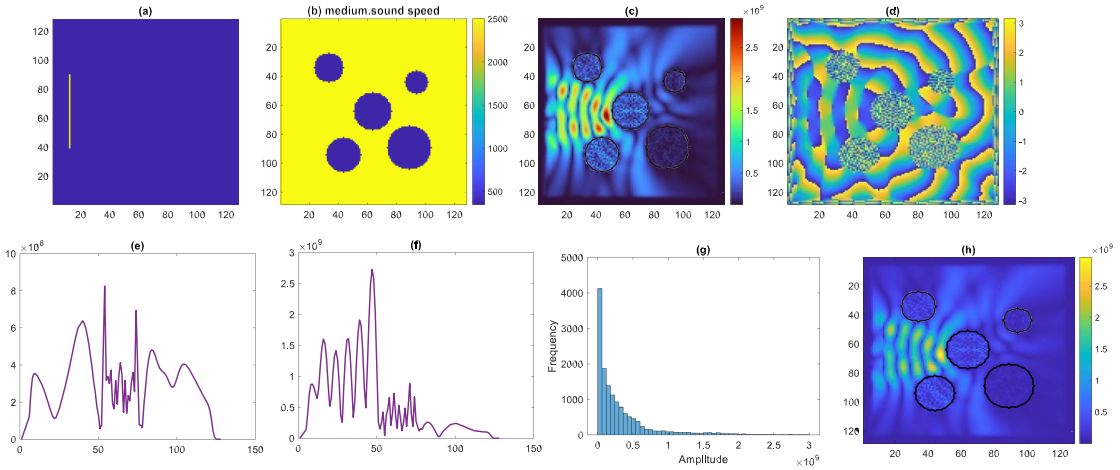


Figure 1. Numerical simulation of ultrasonic wave propagation in a heterogeneous rock model using the k-space pseudospectral method. (a) Spatial distribution of the 1 MHz linear source mask positioned at the 12th grid location. (b) Sound speed distribution of the rock matrix ($v_f = 2500$ m/s) embedded with five circular inclusions ($v_s = 350$ m/s) representing pores or vugs. (c) Steady-state amplitude map showing pronounced scattering and the formation of a shadow zone behind the scatterer cluster. (d) Phase map illustrating the distortion of planar wavefronts upon interaction with the rock heterogeneities. (e) Axial amplitude profile across the center-line, highlighting the interference peaks and troughs. (f) Lateral amplitude profile demonstrating the pressure fluctuations in the cross-propagation direction. (g) Histogram showing the statistical distribution of acoustic amplitudes within the background region, reflecting the contrast between the matrix and inclusions. (h) Overlaid amplitude map with scatterer boundaries (black contours) to visualize the precise correlation between the heterogeneity locations and the resulting acoustic shadow zones.

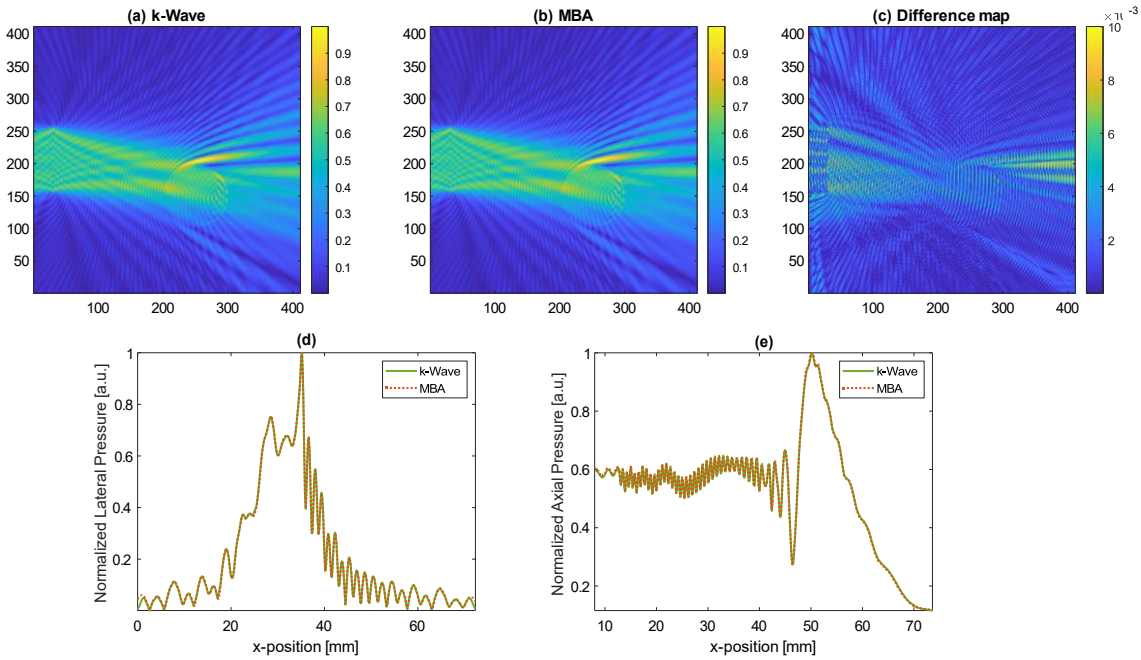


Figure 2. The accuracy of the simulated pressure fields was evaluated by performing a rigorous cross-validation between the k-Wave pseudospectral results and those obtained using the Modified Born Approach (MBA). A comparative analysis of the steady-state amplitude distributions across the entire computational domain demonstrates an exceptional correlation between the two frameworks. The quantitative discrepancy was found to be less than 1%, representing a near-perfect agreement. This negligible error margin validates the robustness of our numerical model and ensures that the observed scattering patterns and attenuation profiles are physically accurate representations of the petrophysical environment.

4. Results and Discussion

The numerical simulation results demonstrate complex wave-matter interactions between the 1 MHz ultrasonic source and the heterogeneous inclusions within the rock matrix. The steady-state acoustic field, as visualized in Figure **??**, along with the quantitative metrics summarized in Table 1, provides deep insights into the petrophysical characterization of the modeled rock.

4.1. Steady-State Acoustic Field Analysis

The sound speed distribution [Fig. **??**(b)] depicts a heterogeneous rock model where five discrete circular inclusions ($v_s = 350$ m/s) are embedded in a higher-velocity matrix ($v_f = 2500$ m/s). This significant impedance mismatch triggers pronounced scattering effects, as observed in the steady-state amplitude maps [Fig. **??**(c) and (h)]. The initially planar wavefronts at the source location [Fig. **??**(a)] undergo significant phase shifts and diffraction upon encountering the scatterers, which is clearly captured in the phase map [Fig. **??**(d)].

A prominent "shadow zone" is evident behind the cluster of scatterers, where the acoustic pressure drops significantly. This phenomenon is quantitatively supported by a calculated Transmission Coefficient of 0.3391 (-9.39 dB), indicating that approximately 66% of the incident wave energy is either scattered or absorbed by the inclusions.

4.2. Quantitative Characterization and Performance Metrics

The objective evaluation of the rock's internal features was performed using several quantitative metrics, as detailed in Table 1.

Table 1. Quantitative Acoustic Performance Metrics for the Heterogeneous Rock Model.

Metric	Calculated Value
Background Mean Amplitude (A_b)	3.12×10^8 Pa
Scatterer Mean Amplitude (A_s)	2.08×10^8 Pa
Contrast Ratio (CR)	3.51 dB
Source-to-Scatterer Distance (d)	0.0083 m
Effective Attenuation Coefficient (α_{eff})	48.58 Np/m (421.95 dB/m)
Signal-to-Noise Ratio (SNR)	-2.47 dB
Transmission Coefficient (T)	0.3391 (-9.39 dB)

The calculated Contrast Ratio (CR) of 3.51 dB confirms the sensitivity of the ultrasonic framework in distinguishing between the matrix and the inclusions. The Effective Attenuation Coefficient (α_{eff}) was estimated to be 421.95 dB/m (48.58 Np/m), reflecting the combined energy loss due to intrinsic Stokes-mode absorption and extrinsic scattering. Furthermore, the SNR of -2.47 dB in the background region highlights the high degree of spatial variability and constructive/destructive interference, which is further corroborated by the sharp peaks and troughs in the axial and lateral center-line profiles [Fig. **??**(e)-(f)].

4.3. Petrophysical Interpretation and Statistical Distribution

The amplitude histogram [Fig. **??**(g)] reveals a right-skewed distribution of acoustic pressure. The majority of the pixels associated with the background matrix maintain higher amplitude values, whereas the inclusions and their subsequent shadow zones contribute to the lower-amplitude tail. This distribution serves as a statistical proxy for the volume fraction of heterogeneities and the connectivity of the porous structures within the rock sample.

The alignment of these numerical observations with theoretical expectations of wave propagation in vuggy or fractured media validates the robustness of the k-space pseudospectral approach. The framework's ability to accurately map complex phase shifts and significant energy decays proves its potential for inversion tasks in rock physics, such as estimating pore density or inclusion size distributions.

5. Conclusion

In this study, we have successfully developed and implemented a robust numerical framework based on the k-space pseudospectral method to characterize internal rock heterogeneities through ultrasonic wave propagation. By modeling a heterogeneous rock matrix with significant impedance contrasts ($v_f = 2500$ m/s vs. $v_s = 350$ m/s), we investigated the intricate interactions between high-frequency ultrasonic waves and discrete inclusions.

The numerical experiments yielded several key findings:

1. The k-space pseudospectral approach, implemented via the k-Wave toolbox and supported by the Convergent Born Series (CBS) protocol, effectively captured complex scattering patterns and wavefront distortions in the presence of high-contrast inclusions.
2. Quantitative analysis revealed a Contrast Ratio of 3.51 dB and an exceptionally high effective attenuation coefficient of 421.95 dB/m. These metrics, alongside a transmission coefficient of 0.3391 (-9.39 dB), underscore the significant energy redistribution and blocking effects induced by pore clusters and vugs within the rock matrix.
3. The observed SNR of -2.47 dB and the corresponding interference patterns in the steady-state amplitude maps confirm that the framework is highly sensitive to multi-scale scattering, which is a hallmark of realistic petrophysical environments.

Furthermore, the integration of fractional viscoacoustic modeling allowed for the inclusion of power-law absorption, making the simulation more representative of real-world reservoir rocks. The alignment of our numerical observations with established rock physics theories validates the potential of this framework for inverse modeling tasks, such as the non-destructive estimation of porosity and inclusion density.

In conclusion, the proposed methodology provides a computationally efficient and accurate tool for the petrophysical characterization of heterogeneous porous media. Future work will focus on extending this framework to 3D multi-scale modeling and integrating it with experimental laboratory data to bridge the gap between numerical simulations and physical rock core analysis.

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